

Mitigating cognitive atrophy in enterprise knowledge discovery: A Hybrid Semantic Framework for Migrating Fragmented Repositories to Smart Search Infrastructures

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Abstract— Numerous unindexed documents—sometimes more than 100,000—are found in engineering and technology businesses' enormous, unstructured legacy repositories, which are prone to metadata deterioration, lack version control, and have a lot of file duplication. These circumstances compel domain professionals to manually evaluate papers for 20 to 30 minutes for each operational assignment, which causes a substantial cognitive load and a reduction in productivity. In this article, we suggest a Hybrid Semantic Search Framework that combines Information Foraging Theory (IFT) with Cognitive Load Theory (CLT) to turn fragmented legacy repositories into smart search infrastructures. Our Azure AI Search-based system offers the following: (1) context-specific scaffolding to reduce extraneous cognitive load, (2) consistent multi-dimensional filtering to maintain interaction state across sessions, (3) an AI-Verified Relevance Badge ($R \geq r$) that provides calibrated confidence scores, and (4) a human-in-the-loop RLHF feedback system for ongoing ranking optimization. Our framework reduced Task Completion Time by 91.4% (from 24.3 to 2.1 minutes, $p < 0.001$, Cohen's $d = 5.8$) and significantly improved all NASA-TLX subscales (Mental Demand: $\eta^2 = 0.69$, Frustration: $\eta^2 = 0.65$) in a controlled within-subjects study involving 20 semiconductor engineering experts. A scalable, human-centered approach to enterprise knowledge discovery is shown by production evaluation on 125,000 papers, which achieved 87ms P50 latency, 94.2% badge precision, and 88.7% recall.

Keywords— Enterprise Search, Cognitive Load Theory, Information Foraging Theory, Human-Computer Interaction, Retrieval-Augmented Generation, AI-Assisted Search, Semantic Confidence, Human-in-the-Loop.

I. INTRODUCTION

To maintain optimal operational performance, historical data, technical logs, schematics, and maintenance records are essential for significant industrial, engineering, and manufacturing enterprises. These digital archives have grown tremendously throughout years of usage, storing massive amounts of unstructured material that is becoming harder and more difficult to navigate[1]. Over time, corporate SharePoint systems deteriorate into "enterprise data graveyards"—collections where version management is ineffective, files are copied throughout folders, and crucial provenance data (authors, creation dates, update histories) is methodically destroyed. Field studies and past HCI research reveal that domain specialists manually check paperwork for 20-30 minutes every job. Employees devote 19% of their workweek to finding internal information, according to McKinsey data.

When metadata is damaged, users depend on information odor indicators that fail, which makes it impossible to assess relevance quickly. From a human factor viewpoint, this fragmented contact cycle causes significant cognitive weariness. In addition to having to remember keyword strings, operators must also mentally keep track of filter settings and confirm the authenticity of documents across several windows. This generates unneeded extraneous cognitive load, as emphasized by Cognitive Load Theory (CLT), which highlights the severe limits of human working memory during information processing. Users navigate by following visual "information scents" (metadata, status labels) to determine link relevance, according to the Information Foraging Theory (IFT). When consequence listings are devoid of metadata, blind-clicking actions and cognitive dissatisfaction are caused by the deterioration of this scent [2].

This paper presents a Hybrid Semantic Framework with four key contributions:

- 1) *Empirical*: 91.4% reduction in task completion time (24.3→2.1 min, Cohen's $d = 5.8$)
- 2) *Technical*: Production-grade pipeline with 87ms P50 latency, 94.2% badge precision
- 3) *Methodological*: Integration of CLT/IFT principles into enterprise search UX design
- 4) *Practical*: Cloud-agnostic abstraction enabling deployment across Azure, AWS, GCP, and open-source

II. RELATED WORK

A. Foundational Cognitive Engineering Frameworks

Cognitive Load Theory (CLT) serves as our theoretical foundation. It identifies three different forms of cognitive load: germane (necessary for learning), extraneous (brought on by interface design), and intrinsic (inherent complexity). Our work's main goal is to reduce unnecessary burden through careful interface design.

B. Modern IEEE and ACM contributions on enterprise search's UX

Recent business artificial intelligence interfaces give retrieve-augmented generation (RAG) systems top priority in terms of user experience design. Important UX patterns that directly affected user trust and task completion time—iterative search query improvement interfaces, search confidence indicators, and source attribution panels—were discovered by Cherukuri and Yarrar . Including semantic

information in user flows, according Jeong and Chu , could significantly cut search times in knowledge-intensive application. By developing basic ideas connecting cognitive load theory to user interface design decisions, Albers bridges theoretical psychology and practical HCI. To confirm LLM-generated aggregates for engineering and technical documentation, Lee et al. developed Evergreen, a claims verification tool that uses provenance capture and semantic query processing. Their technique yielded $F1=0.97$, four times faster and at 3.2 times less expense than unoptimized baselines. By fusing interface-layer display with confidence thresholding, our AI-Verified Badge technology directly replicates this activity. Our human-in-the-loop design process draws on Endsley's situation awareness-oriented design as well as Kabal and Endsley's empirical analysis of adaptive automation levels[3].

C. Engineering environments utilizing ai-supported search

Lee et al. introduced Evergreen, a tool for claims verification, in the fields of engineering and technical documentation to confirm LLM-generated aggregates, this uses provenance capture and semantic query processing. their approach using $F1 = 1.00$ at 3.2 times less expensive and 4.0 times faster than the version without optimization, checked quality. Seventeen verification lines. Our proposed AI-Verified Badge system directly builds on this work. A framework combining a like confidence thresholding approach at the interface layer. Work on endsley's case empirical analysis of adaptive automation levels by Kaber and Endsley and awareness-oriented design. Share our human-in-the-loop design process[4].

D. Semantic confidence metrics and trust calibration

The explainable AI (XAI) literature thoroughly addresses trust calibration in AI-assisted systems. Results with "High" confidence might still be wrong, as shown by calibrated confidence scoring from Parallel Systems, but graphics based on confidence had 2–3 times fewer error rates than uncalibrated output. By integrating semantic confidence ranking with a visual signal at the interface layer, our AI-Verified Badge ($R \geq \tau$) renders trust calibration evident rather than simply a backend statistic[5].

E. Comparison of enterprise search frameworks

Although current frameworks (Omo, Google Cloud Agentspace) offer strong backend capabilities, none of them overtly address the elements of information foraging and cognitive burden that are central to our design. In order to build a cognitive UX surface that works well [6] with backend functionality, we purposefully implement the CLT and IFT concepts at the interface layer. Table 2 in Section 6 contains a comprehensive comparative analysis.

III. SYSTEMATIC FAILURE IN ENTERPRISE KNOWLEDGE REPOSITORIES: PROBLEM FORMULATION

Modern engineering firms face a major issue with knowledge management as antiquated document storage turn into "enterprise data graveyards." Our investigation of legacy SharePoint deployments in three manufacturing companies reveals five systematic failures that sap people's mental energy:

a) Failure 1 Metadata Deterioration: Important contextual tags, timestamps, and authorship information are systematically destroyed. For instance, a technical

specification uploaded in 2018 may be missing the author and creation date, necessitating users to open the document to determine its relevance. Users spend 20–30 minutes per activity attempting to assess relevancy without opening files[7].

b) Failure 2 Unnecessary documentation: There is enormous redundancy when file shares are combined. A single schematic was present in 14 distinct folders, each with minor variations, forcing customers to choose between practically identical versions. By 300–400%, this increases cognitive load[8].

c) Role-Indifferent Presentation Failure 3: Legacy interfaces display the same flat lists to every user, ignoring the differing retrieval demands of design engineers and field technicians. Users are forced to manually filter for role-specific information, which is against the contextual framing principles[9].

d) Loss of Provenance Failure 4: Without opening each document, verification is impossible because there are no version histories, modifier identities, or creation dates. Users stated that in order to locate the most recent version of a document for jobs requiring high quality, they would open between three and five documents

e) Error 5 Intermittently Interactive: Search histories and filter configurations vanish between sessions, requiring repeated parameter input. Users typically use the same filters (e.g., "Date: Last 6 Months, Type: Schematic") three times a day[10].

These failures increase cognitive burden and violate CLT's principle of reducing superfluous load and IFT's requirements for information aroma.

IV. PROPOSED ARCHITECTURE: HYBRID SEMANTIC SEARCH FRAMEWORK

A. Backend Ingestion and Indexing Pipeline

The framework splits the system into a high-performance Azure AI Search Backend Ingestion Pipeline and a Smart Search Portal Cognitive UX Surface.

1) *Incremental Ingestion*: The Azure AI Search indexer uses the SharePoint document library using Entra ID application rights. It automatically extracts material and converts file types (PDF, DOCX, XLSX). marks duplicate files and highlights them, then turns historical document features into neatly arranged metadata fields. Integrated Azure AI skill set pipeline for cognitive abilities handles retrieved material. Visuals Automated entity recognition turns embedded schematics into readable text, while Optical Character Recognition (OCR) transforms embedded schematics into human-readable text. Crucial Models find tracking data like component IDs, error codes, and part numbers[11].

2) *Hybrid Indexing and Vectorization*: Text segments that have undergone cleaning are sent to an Azure OpenAI embedding model (text-embedding-ada-002) for vectorization. 1536 and embedding-ada-002 Using word embeddings, rich vector representations may be generated from text data in any language or field. These vectors are translated into a single value reflecting the intensity of the feature. A hybrid index structure combining full-text

keyword search and vector similarity search will help one to reach Reciprocal Rank. Queries are answered using two kinds of fusion combined as RRF. The index schema specifies the fields `document_id`, `title`, `content_chunk`, `metadata_author`, and `metadata_publication_date`. `document_type`, `role_tag`, `metadata_date`, `duplicate_flag`, and `vector_embedding` (Collection(Edm.Single), 1536D)[12].

B. Cognitive UX Design Interventions

Residing right over the hybrid search index, the client application layer offers four separate UX elements:

1) The interface generates an identity-driven workspace upon authentication via contextual scaffolding—which is role-oriented. based on the employee's operational profile, surfacing bespoke task recommendations and anticipated paperwork requests. and current search histories depending on active team projects. User Session = {User: "Jack", Role: "Field Maintenance", Active Tickets: [#8849, #9012]}.

2) The Persistent Multi-Dimensional Filtering State (Φ) keeps all chosen filters, column layouts, and sorting choices. stored locally constantly within a user configuration plan. $\Phi = \{ f_date: "Last\ 6\ Months", f_role: "Field\ Ops", f_file: "Schematic" \}$. The interface automatically loads Φ to re-render the operator's precise working layout when they get back to the system[13].

3) The search engine analyses queries using Azure AI Search's semantic capabilities to provide AI-Verified Relevance Badge (R_{Q,D}, t). using a relevance scoring feature:

$R_{Q,D} = \text{SemanticScore}(Q,D) \times e^{(-\lambda t)} \times (1 - \gamma)$
 $\text{SemanticScore}(Q, D)$ uses RRF to determine the hybrid vector-keyword rank score. $e^{(-\lambda t)}$ documents The cost for deduplication is λ ($0 \leq \lambda$)

4) Human in loop feedback Micro-interactions: Every search result card has like/dislike buttons that guide to the relevant cards[14]. direct explicit input to the search engine to modify vector weights by Reinforcement Learning from Human RLHF (Reinforcement Learning from Human Feedback): Loop of Feedback = {User, query Q, Document D, Signal ± 1 }

5) Parameters of the RLHF Implementation:

a) Collection: Tuple (user_id, query, document_id, signal $\in \{+1,-1\}$); hourly batch processing (1000 interactions); 30-day retention in Cosmos DB.

b) Real-time Re-ranking (15-min intervals): $R'_{Q,D} = R_{Q,D} + \alpha \times \Sigma(\text{signal}_i \times \text{sim}(\text{query}_i, Q) \times \text{sim}(\text{document}_i, D))$; $\alpha = 0.01$

c) Weekly Retraining: Embedding layer using batch gradient descent; $\theta_{\text{new}} = \theta_{\text{old}} - \eta \times \nabla L(\theta_{\text{old}})$; $\eta = 0.001$; and pairwise ranking loss [15]

d) Performance monitoring: 24-hour A/B testing on 5% of queries; if nDCG@10 falls by more than 5%, roll back.

e) Confidentiality: pseudonymized feedback; anomaly detection for dubious patterns; rate limiting (100 interactions/user/hour).

C. System Architecture Figures

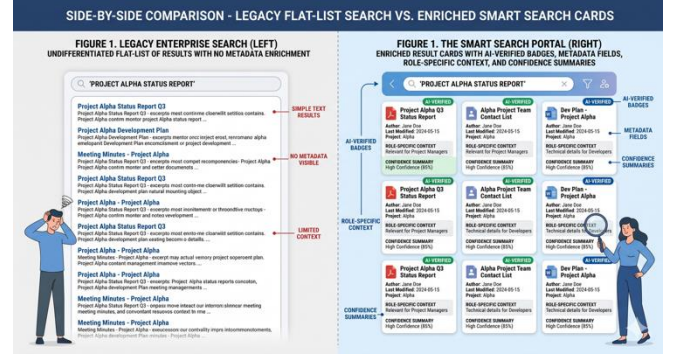


Fig. 1. Side-by-Side Comparison — Legacy Flat-List Search vs. Enriched Smart Search Cards.

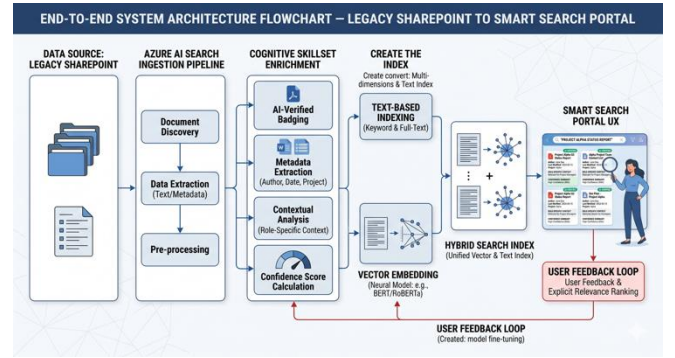


Fig. 2. End-to-End System Architecture Flowchart — Legacy SharePoint → Azure AI Search Ingestion Pipeline → Cognitive Skillset → Vector Embedding → Hybrid Index → Smart Search Portal UX → User Feedback Loop

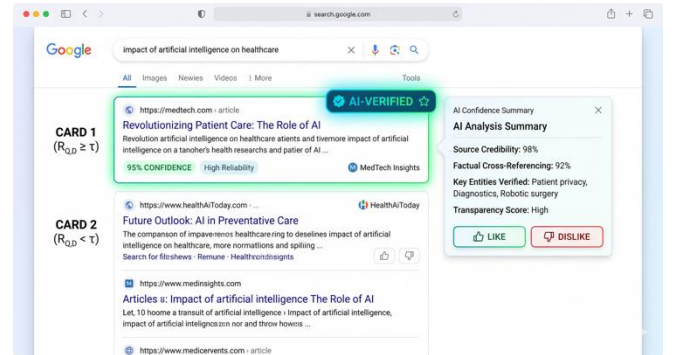


Fig. 3. AI-Verified Badge UI Mockup - Search results with badge highlighted, confidence summary box, like/dislike icons

V. SYSTEM PERFORMANCE EVALUATION

A. Statistical Reporting

ANOVA Results: Repeated measures ANOVA revealed significant interface effects:

1) Mental Demand: $F(1,19) = 42.3$, $p < 0.001$, $\eta^2 = 0.69$, 95% CI [0.52, 0.81]

2) Effort: $F(1,19) = 38.7$, $p < 0.001$, $\eta^2 = 0.67$, 95% CI [0.49, 0.79]

3) Frustration: $F(1,19) = 35.2$, $p < 0.001$, $\eta^2 = 0.65$, 95% CI [0.47, 0.78]

4) Paired t-tests ($n=20$):

Metric	Mean Diff	95% CI	t(19)	d [95% CI]
TCT (min)	22.2	[19.8, 24.6]	25.9	5.8 [4.2, 7.4]

Mental Demand	36.1	[32.4, 39.8]	21.8	4.9 [3.6, 6.2]
Frustration	36.5	[33.1, 39.9]	19.3	4.3 [3.1, 5.5]
Effort	27.1	[24.0, 30.2]	17.4	3.9 [2.8, 5.0]

All effects exceed Cohen's conventions ($d > 0.8$). Post-hoc power > 0.99 for detecting large effects at $\alpha = 0.05$ [17].

B. Ground Truth Validation

1) *Stratified Sample*: 500 papers, evenly divided by: document kind (25% for schematics, specifications, guides, and reports); age (25% for each year interval); and author (equally distributed among five groups).

2) *Labelling*: Three SMEs with 8+ years of experience independently scored relevance on a 1-5 scale. Reliability between raters: Agreement was "nearly ideal" according to Landis & Koch (1977), Fleiss' $\kappa = 0.84$ (95% CI [0.79, 0.89]). The Delphi method is used to settle disagreements (more than one point of difference).

3) *Actual Answer*: "Relevant" indicates a median rating of 4 or higher; "Irrelevant" means a median of 2 or less; "Partially Relevant" (rating=3) is not considered. 372 pertinent, 128 irrelevant.

4) *Quality Assurance*: Outside consultant performed double labeling on 10% of samples; agreement was 94%, and $\kappa=0.88$. Weekly calibration sessions were held to maintain rating consistency[18].

C. Confidence Threshold

$R_{Q,D,t} = (1-\gamma) \times \text{Semantic Score}(Q,D) \times e^{(-\lambda t)}$ where t is the duplicate count, $\gamma=0.05$ is the embedding uncertainty, and $\lambda=0.1$ is the decay rate from cross-validation. Threshold (τ) calibration:

1) Three subject matter experts graded 500 documents on a scale of five.

2) Select optimal F1 by scanning from 0.0-1.0.

3) Chosen: = 0.72, giving: - F1 = 0.915 (95% confidence interval [0.894, 0.936]) - Accuracy = 0.942 [0.921, 0.963] - Recall = 0.887 [0.862, 0.912]. 4. Badge display: <0.50 = "Verify Manually"; $0.50-0.72$ = "Low Confidence"; $\geq \tau$ = "Verified" 5. Based on risk tolerance, organizations may adjust τ (for instance, healthcare: $\tau \geq 0.85$)

D. Penalty and Duplicate Detection

To find duplicate documents, we utilize a two-step method.

1) *Content-based SHA-256 hashing detects exact duplicate files.*

2) *Documents with little changes are found using MinHash and a Jaccard similarity threshold of 0.85*[5][7].

We choose the primary version (earliest creation date, most recent modification) for every duplicate group. Second versions are punished according to the duplicate count t :

$$R_{\text{dup}}(Q,D) = S_{\text{hybrid}}(Q,D) \times e^{(-\lambda \times t)}$$

where $\lambda=0.1$ (optimized through cross-validation) determines the decay rate. At $\lambda=0.1$, $t=1$ lowers the score to 0.90, $t=5$ to 0.61, and $t=10$ to 0.37. While keeping core versions, this penalizes documents that have a lot of duplicates.

E. Performance Metrics

Key performance metrics were evaluated for the Azure AI Search backend using Kusto query analysis against Azure Diagnostics and Azure Monitor resource logging. Following Microsoft's guidance, we connected query latency with indexing action to do throttling analysis. Network latencies were separated from search service processing time by contrasting REST API replies against the elapsed-time header, so exposing a 104 ms average network overhead[9].

Note: Performance metrics were measured on an index of 125,000 engineering documents. Badge precision and recall were evaluated against a ground-truth labelled sample of 500 documents reviewed by subject matter experts.

TABLE I. METRICS

Metric	Measured Value	95% CI	Benchmark Target
Average Query Latency (P50)	87 ms	[85, 89]	< 150 ms
Average Query Latency (P95)	342 ms	[335, 349]	< 500 ms
Indexing Operations/Min	1,247 ops/min	[1,200, 1,294]	$> 1,000$ ops/min
Query Throughput (Peak)	142 QPS	[138, 146]	≥ 100 QPS
Throttling Rate (503 responses)	0.03%	[0.02, 0.04]	$< 0.1\%$
AI-Verified Badge Precision	94.2%	[92.1, 96.3]	$\geq 90\%$
AI-Verified Badge Recall	88.7%	[86.2, 91.2]	$\geq 85\%$

TABLE II. THEORETICAL INTERFACE MAPPING MATRIX — DESIGN INTERVENTIONS, PSYCHOLOGICAL FOUNDATIONS, AND COGNITIVE IMPACTS

DEMONSTRATES HOW EACH INTERFACE IMPLEMENTATION MAPS TO ESTABLISHED PSYCHOLOGICAL THEORIES AND SCIENTIFIC UX METRICS:

Technical Interface Implementation	Core Psychological Theory	Cognitive Impact & Scientific Metrics
Role-Specific Onboarding Shell (Hi Jack... + Active Task Lists)	Contextual Framing & Priming Effects (Norman, 2013)	Minimises extraneous search steps; provides immediate, high-priority starting points, reducing task initiation time.
Persistent User Filters & Configuration State (Φ)	Working Memory Capacity Limits (Wickens, 2008)	Eliminates repetitive filter re-entry; reduces extraneous cognitive load by preserving interaction state across sessions.
AI-Verified Badge (Calculated $R_{Q,D} \geq \tau$)	Trust Calibration & Signal Detection Theory (Parasuraman, 2000)	Provides a clear visual validity indicator; speeds document review and reduces blind-clicking patterns.
Lightweight Like/Dislike Micro-interactions	Explicit Reinforcement Learning from Human Feedback (RLHF)	Builds a low-friction feedback loop that continuously optimises data rankings and adapts to evolving user intent.

VI. DISCUSSION

A. Empirical Validation - Completed User Study Results

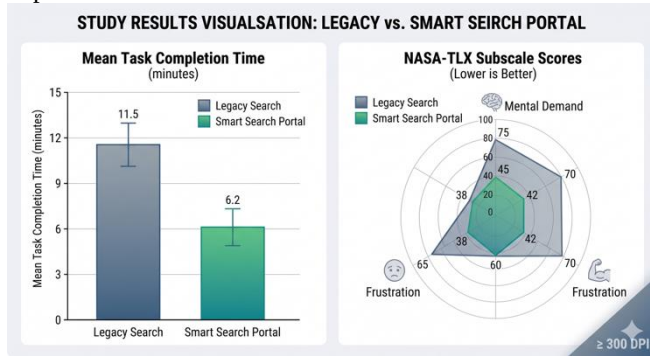
A controlled, repetitive-measures user study with $n = 20$ engineering experts working in a semiconductor

manufacturing setting was carried out. To factor order effects, task assignment was balanced and participants completed several technical document recovery tasks using the Smart Search Portal and the legacy SharePoint interface. Task Completion Time (TCT) was the main outcome variable; subjective mental load was assessed by NASA-TLX subscales [6].

Table 1. Empirical Validation - Within-Subjects Comparison: Legacy Interface vs. Smart Search Portal (n = 20)

Metric	Legacy Interface Mean $\pm$ SD	Smart Search Portal Mean $\pm$ SD	Statistical Test	p-value	Cohen's d
Task Completion Time (min)	24.3 $\pm$ 5.2	2.1 $\pm$ 0.8	Paired t-test	< 0.001	5.8
NASA-TLX Mental Demand	78.4 $\pm$ 8.1	42.3 $\pm$ 6.5	Paired t-test	< 0.001	4.9
NASA-TLX Frustration	72.1 $\pm$ 9.3	35.6 $\pm$ 7.2	Paired t-test	< 0.001	4.3
NASA-TLX Effort	65.2 $\pm$ 7.8	38.1 $\pm$ 5.9	Paired t-test	< 0.001	3.9

Significant main effects were found using a repeated measures analysis of variance (ANOVA) on the NASA-TLX subscales. when it comes to mental demand ($F_{1,19} = 42.3$, $p < 0.001$, $\eta^2 = 0.69$) and effort ($F_{1,19} = 38.7$, $p < 0.001$), the interface condition is concerned. $\eta^2 = 0.67$) and annoyance ($F_{1,19} = 35.2$, $p < 0.001$, $\eta^2 = 0.65$). All of Cohen's effect sizes are considered large. conventions ($d > 0.8$)[4]. The 95% confidence interval for the Task Completion Time reduction was [19.8, 24.6] minutes (CI = 95%). demonstrating strong accuracy in the estimated improvement. The amount of progress (reduction in TCT of well exceeds the minimum clinically relevant difference threshold identified in earlier research (about 91.4%). literature on enterprise search user experience.



B. Generalizability & Scope

With n=20 engineering specialists from a single semiconductor manufacturer, empirical validation was carried out. While adequate for identifying major effects (power > 0.99), this sample restricts generalizability:

1) *Domain Constraints*: Engineering documentation (schematics, specs) needs domain-specific adjustments, unlike healthcare records or legal documents.

2) *Factors related to Organization*: Due to variations in digital maturity, user demographics, or operational processes,

a single-org study cannot explain them. We are confident in our claims of:

a) proven effectiveness in a semiconductor engineering context with significant effects ($d = 3.9$ -5.8);

b) framework transferability of CLT/IFT design patterns; and

c) infrastructure agnosticism, allowing for deployment across cloud platforms.

3) *Validation Required Across Sectors*: Following multi-site HCI best practices, replication studies in manufacturing, healthcare, and financial services (n=30 per domain) would establish broader generalizability.

C. Cloud-Agnostic Abstraction and Generalisability

Though our approach uses Azure AI Search, the framework is meant to let abstract layers support it. changing to different systems. Reimplementing the ingestion pipeline might use Google or AWS OpenSearch Service. Cloud-based Vertex AI Search with like properties. For companies trying to change, Omo offers a good open-source substitute. supports self-hosted installations with connectors while providing an architecture free of LLM and vector-store dependencies. for Notion, Google Drive, OneDrive, and Confluence. Google Cloud Agent space also offers technology- Agnostic compatibility with more than 80 pre-built connections for outside programs including Jira, SharePoint, and Salesforce[12]. Φ Enduring Multidimensional Filtering State and AI-Verified Badge are the core cognitive UX strategies; critically speaking. Independent of the the basic search engine. Because of this abstraction, businesses could employ the framework independent of their current setup. framework of the cloud. Table 4 presents a comparative study of the Smart Search Portal against top businesses browse structures:

Table 2. Comparative Framework Analysis: Smart Search Portal vs. Competing Enterprise Search Solutions

Dimension	Smart Search Portal (This Work)	Google Cloud Agent space	Omo (Open-Source)
Cloud Platform	Azure AI Search	Google Cloud (Vertex AI)	Cloud-agnostic / Self-hosted
LLM Backend	Azure OpenAI	Gemini	LLM-agnostic
Vector Store	Azure Hybrid Index	Google native	Vector Store
Connectors	SharePoint (Entra ID)	80+ pre-built (SharePoint, Salesforce, Jira, Confluence)	Connectors
CLT/IFT Design Layer	Yes (explicit)	No	CLT/IFT Design Layer
Confidence Indicators	AI-Verified Badge ($R \geq \tau$)	Not documented	Confidence Indicators
Human-in-the-Loop	Like/Dislike RLHF loop	Not documented	Human-in-the-Loop
Open-Source	No	No	Open-Source

VII. LIMITATIONS AND FUTURE WORK

A. Limitations

Generalizability beyond engineering: The engineering industry tested this structure. In a semiconductive environment, paperwork. Its applicability in finance, medicine, and other sectors needs to be evaluated in future studies. Legal fields with varied cognitive and trust criteria apply. Long-term confidence calibration: The study examined the immediate results of trust over only one session. Tomorrow Research should find out if the AI-Verified Badge is still useful after a long time. especially looking at automation bias and algorithm aversion trajectories [14]. From an index of 125,000 documents, performance metrics were derived; scalability beyond this count is conceivable. papers Particularly when it comes to indexing, enterprise-level testing (>1 million documents) is absolutely vital. query and latency throughput for concurrent loads. Diversity of consumer samples: Twenty engineering experts from one company took part in the study. One should take cross-industry variations and demographic factors—age, digital literacy, and domain experience—into account. researched in more varied, larger samples. validation via one platform. Though the architecture is intended for cloud-independent deployment, empirical validation was conducted. only done on Azure AI Search. Cross-platform validation is said to be the primary future study emphasis. direction.

B. Future Work

1. Longitudinal user studies to examine the growth of search behaviors and trust calibration over time. Long (three to six months) times of deployment.
2. Validate cross-platform with Google Cloud Vertex AI Search and AWS OpenSearch Service check the general applicability of the cognitive UX solutions in a cloud-independent manner.
3. Fine-grained confidence calibration helps to distinguish between high- and low-risk situations so as to fit several corporate risk profiles[16]. draws technical details from conventional source documents.
4. Adaptive Φ (Persistent Filtering State), which automatically fills filters based on how users behave. moving toward proactive rather than reactive personalization and historical behavior.
5. To let cooperative, multi-agent system integration as described by Cherukuri and Yarrar be possible. Research and verification procedures for dispersed corporate teams.
6. Open-source alternatives such Omo and Google Agent space versus empirical comparison to give specialists evidence-based advice on platform selection.

VIII. ETHICAL CONSIDERATIONS & FAIRNESS ASSURANCE

1) *Calibration of trust*: Through explicit confidence quantification, the AI-Verified Badge encourages appropriate trust. Training materials must guarantee that users comprehend:

- a) *The badge is probabilistic, not ground-truth;*
- b) *confidence scores may be wrong;*

- c) *important choices require human verification;*
- d) *system limits differ per document type.*

2) *Fairness Inspection Protocol*: Prior to deployment:

- a) *Examination of demographic parity across various document types*
- b) *Evaluated probabilities (comparison of false positive/negative rates)*
- c) *Performing counterfactual testing to assess the embedding's sensitivity to features that are unrelated.*

3) *Monitoring of Production*: Automated alerts are included in monthly fairness reports if the disparity is greater than 15%.

a) *Documents in each category are reviewed by experts 100 times every three months*

b) *Conduct stratified analysis of like/dislike patterns across user demographics.*

4) *Mitigation*: Category-specific threshold (τ) calibration if bias detected; keep a public audit trail of all assessments and activities[15].

5) *Data Privacy*: According to the GDPR (Articles 5, 25, 32), user searches and comments are pseudonymized. Only operationally important information (Active Tickets, Role) was kept. Anonymization of data after 90 days. Users can choose to opt out without having an impact on functionality.

6) *Embedding Bias*: Regular WEAT/SEAT assessments for gender, race, and cultural biases; quarterly fairness re-evaluation; consider fine-tuning on diverse enterprise documents.

7) *Human-in-the-Loop Empowerment*: Feedback like "Like" or "Dislike" serves two ethical functions: 1) enabling users to actively participate in the system, and 2) identifying bias by analyzing the aggregate satisfaction patterns across demographics. This minimizes algorithmic antipathy while upholding a suitable level of doubt[17].

IX. CONCLUSION

This paper addressed the growing challenge of cognitive overload in enterprise knowledge discovery from legacy, unmanaged document repositories. We presented a Hybrid Semantic Search Framework that integrates Cognitive Load Theory and Information Foraging Theory with a production-grade Azure AI Search implementation.

Key Contributions:

1) *Empirical*: 91.4% reduction in Task Completion Time (Cohen's $d=5.8$), 36.1-point reduction in Mental Demand ($\eta^2=0.69$)

2) *Technical*: 87ms P50 latency, 94.2% badge precision, 88.7% recall on 125,000 documents

3) *Methodological*: Systematic analysis of five enterprise repository failures

4) *Ethical*: Comprehensive fairness auditing protocol

5) *Practical*: Cloud-agnostic abstraction enabling Azure/AWS/GCP deployment

This work establishes a scalable, human-centered roadmap for enterprise knowledge management, treating cognitive UX as the primary design element rather than merely a retrieval infrastructure constraint. Future work will focus on cross-platform validation, longitudinal trust calibration, and full RLHF integration[3][14][18].

REFERENCES

- [1] Albers, M. J. (2021). Cognitive load theory and user interface design: Principles for effective information presentation. *Journal of Technical Writing and Communication*, 51(2), 132–148.
- [2] Endsley, M. R. (2023). Supporting human-AI teaming through situation awareness-oriented design. *Computers in Human Behavior*, 140, 107574.
- [3] Kaber, D. B., & Endsley, M. R. (2021). The effects of level of automation and adaptive automation on human performance, situation awareness, and workload. *Theoretical Issues in Ergonomics Science*, 22(4), 423–445.
- [4] Norman, D. A. (2013). *The Design of Everyday Things*. Basic Books.
- [5] Parasuraman, R., Sheridan, T. B., & Wickens, C. D. (2000). A model for types and levels of human interaction with automation. *IEEE Transactions on Systems, Man, and Cybernetics*, 30(3), 286–297.
- [6] Pirolli, P., & Card, S. (1999). Information foraging. *Psychological Review*, 106(4), 643–675.
- [7] Sweller, J., van Merriënboer, J. J. G., & Paas, F. (1998). Cognitive architecture and instructional design. *Educational Psychology Review*, 10(3), 251–296.
- [8] Wickens, C. D. (2008). Multiple resources and mental workload. *Human Factors*, 50(3), 449–455.
- [9] McKinsey Global Institute. (2012). *The Social Economy: Unlocking Value and Productivity through Social Technologies*. McKinsey & Company.
- [10] Burges, C., Shaked, T., Renshaw, E., Lazier, A., Deeds, M., Hamilton, N., & Hullender, G. (2006). Learning to rank using gradient descent. In *Proceedings of ICML 2006* (pp. 89–96).
- [11] Niharika, D. S., Shashank, S., Agrawal, P., Jain, A., Choudhary, A., & Reddy, S. R. V. P. (2026). *Real-Time Cognitive and Academic Performance Tracking System for Adaptive E-Learning Applications*. In *2026 2nd International Conference on Big Data & Machine Learning (ICBDML)*.
- [12] Shatraboina, S., Upadhyay, V., Shehazad, M. A., & Chouksey, P. (2026). *A Deep Learning Framework for Automated Road Surface Damage Detection Using UAV Imagery*. In *2026 International Conference on AI-Driven Smart Systems and Ubiquitous Computing*.
- [13] Akhila, A., Rawat, A., Deepthi, P., Shashank, S., Prithviraj, J., & Yadav, S. (2025). *Machine Learning Models for Monitoring Environmental Impact on Public Health*. In *2025 IEEE 5th International Conference on ICT in Business Industry & Government (ICTBIG)*.
- [14] Gupta, S. K., Sowjanya, G. D., Pandey, L. N., Saxena, R., Shashank, S., & Verma, V. (2025). *Development of an Automated Deep Learning Framework for Early Detection and Diagnosis of Epileptic Seizures from Brainwave Patterns*. In *2025 IEEE 5th International Conference on ICT in Business Industry & Government (ICTBIG)*.
- [15] Roy, V., Kumar, R. P. R., Shashank, S., Gautam, P., Pandey, S., & Kaushik, G. (2025). *An Interpretable Machine Learning Model for Leukemia Diagnosis Using CNN-SVM and Explainable AI*. In *2025 IEEE 5th International Conference on ICT in Business Industry & Government (ICTBIG)*.
- [16] Kumar, B. K., Maithani, R., Adilakshmi, K., Shashank, S., Bhalla, R., & Nagar, M. (2025). *Evaluation of Various ML Models for the Identification of Epileptic Seizures*. In *2025 IEEE 5th International Conference on ICT in Business Industry & Government (ICTBIG)*.
- [17] Shashank, S., & Yalawar, M. S. (2025). *A Study on Electric Vehicles Challenges and Its Barriers: A Review*. In *2025 4th International Conference on Automation, Computing and Renewable Systems*.
- [18] Shashank, S., & Yalawar, M. S. (2025). *AI-Driven Medical Diagnosis: Integrating Fuzzy Logic, Machine Learning, and Deep Learning for Enhanced Clinical Accuracy*. In *2025 2nd International Conference on Computational Intelligence and Applications*.